

# Predictability Comparison of Three Kinds of Robbery Crime Events Using LSTM

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## ABSTRACT

Prediction of crime occurrences is of great significance for underlying criminal risk reduction and public security improvement. In all kinds of crime events, robbery is a more serious and noticeable one that threatens our everyday life. Different kinds of robbery events have different characteristics, and therefore their predictability varies. Previous studies mainly focus on forecast of all kinds of crime events at a fixed spatiotemporal scale, however, special predictability comparison between different robbery events at multiple scales is insufficient. In this paper, we use LSTM to capture long-short space-time dependencies of robbery events, and explore the predictability differences of three kinds of robbery crimes at different spatial and temporal scales. Daily crime occurrence data of Atlanta, US, from 2009 to 2017, and crime data of Baltimore, US, from 2011 to 2016, are respectively used in experiments, and the correlation coefficient between the predicted value and the true value is used to evaluate the model performance. Three robbery events, including commercial robbery, pedestrian robbery, as well as residence robbery, are compared. Experimental results show that the predictability of these three kinds of robbery events differs from each other and varies across different cities. In Atlanta, the commercial robbery is more predictable than that of the residential robbery and pedestrian robbery, with a higher correlation coefficient up to about 0.75. While in Baltimore, the predictability of residential robbery and commercial robbery achieves up to about 0.9, much higher than that of pedestrian robbery. In addition, experiments demonstrate different degree of predictability changes of three robbery events as the spatiotemporal scale changes, reflecting the different geographical distribution characteristics of the three robbery events.

## CCS Concepts

• Computing methodologies → Machine learning → Machine learning approaches → Neural networks • Social and professional topics → User characteristics → Geographic characteristics • Security and privacy → Human and societal aspects of security and privacy → Social aspects of security and privacy

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## Keywords

LSTM; crime prediction; robbery crime comparison; spatiotemporal distribution.

## 1. INTRODUCTION

Violent robbery seriously affects public security and even threatens the mental health of the victims [1]. According to the FBI's annual crime report, there are more than 300,000 robberies in the United States each year. In the past, the police have spent a lot of time and historical data to speculate the crime trend [2], which often leads to the subjective error and makes the police deployment lose the efficiency of crime prevention. Therefore, accurately predicting crimes can greatly prevent the occurrence of crimes and reduce unnecessary manpower expenditure.

With the development of high-performance computer, powerful prediction tools appeared. In particular, the application of machine learning in many fields has brought scientific analytical tools to the prediction methods of crimes. Traditional machine learning algorithms need to build feature engineering for different data sets and cannot obtain a generalized model. Lawrence McClendon used linear Regression and Decision Stump algorithms to predict the crime data and found that the prediction error value of robbery crime was larger than that of murder and rape [9], which indicated that the prediction of robbery crime was more uncertain and challenging. Crime events have strong spatiotemporal dependence and complex category dependence [4], potential crime factors are also influenced by other factors in a short time and a long time. For example, due to the recent robbery in a certain area, the number of robberies in this area may decrease over a certain period of time due to the increased police deployment. Luca Venturini used spectral analysis method and discussed the criminal pattern of space and time, and found that different crime patterns of space and time will be different [10]. Isha Pradhan took the spatiotemporal characteristics of crime into account, and used Random Forest and Decision Tree to build models based on data sets to solve this problem [11], which requires artificial feature engineering. Compared with traditional machine learning algorithms, deep learning tries to learn from experience and understand the world in terms of a hierarchy of concepts, with each concept defined through its relation to simpler [6]. Crime occurs neither uniformly nor randomly in space and time [3]. As a consequence, it is difficult to predict crime trends in different regions. Deep learning has not received sufficient attention in the field of crime prediction. There have been studies using deep learning for forecasting crime through deep neural network (DNN) [14], [15] and convolutional neural network (CNN) [13]. Some researches choose to use the deep learning algorithm to learn the characteristics of crime prediction, and then use the classification algorithm to use the posterior

probability for classification [7]. However, these methods have disadvantages on solving the long-term dependence of crime.

In this paper, we used LSTM to predict crime. LSTM has been applied in many fields. There are many good applications of LSTM in speech recognition, handwriting recognition, text-to-speech synthesis, image captioning, financial market prediction, and fields of spatiotemporal data mining, like environmental pollution monitoring, traffic condition prediction and others. In recent years, it has been found that LSTM can be applied to the field of social sciences, such as criminology [4]. However, existing studies have not compared the predictability of LSTM to different kinds of robbery criminal events. Furthermore, the influence of multi-space-time scales on LSTM for predicting different kinds of robbery crime events is not explored.

This work includes two major contributions. First, we use the LSTM algorithm to predict the daily occurrences of different kinds of robbery crime events across a whole city. Second, we compared the predictability of LSTM for different types of robbery crimes at multiple spatiotemporal scales. The paper is organized as follows: section 2 introduces the data, section 3 describes the LSTM method, section 4 designs the experiments and analyzes the results. Section 5 concludes the paper.

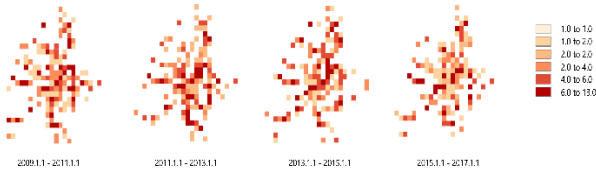
## 2. PRILIMINARIES

This section describes the data, visualizes the data, and describes the data organization.

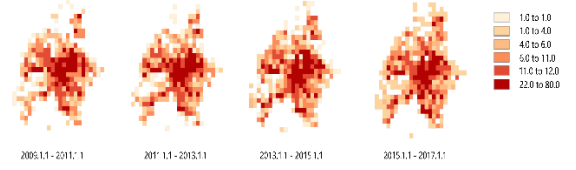
### 2.1 Data Introduction and Visualization

Daily crime data of the city of Atlanta, US, from January 1, 2009 to February 28, 2017, and similar crime data of Baltimore, US, from 2011 to 2016, are used in this paper. Each data set contains the following fields that are used: crime types, time, latitude, and longitude of crime occurrence. There are total 270,686 criminal records in Atlanta, and 285,806 criminal records in Baltimore. Three types of robbery events, including commercial robbery, pedestrian robbery, and residence robbery, are compared and explored.

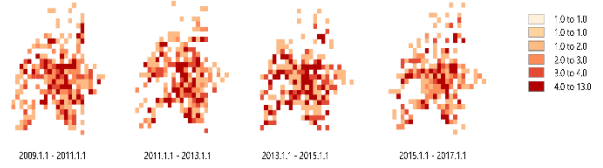
The spatiotemporal distributions of the three robbery crimes in Atlanta are visualized in Fig. 1 (a-c). In the figure, each grid represents the statistical number of crime occurrences within the grid region and the corresponding period. The deeper the color of each grid is, the more crimes occurred in the grid. As shown in the figure, the three types of robbery crimes show certain regularity in time and space, indicating the feasibility of crime prediction. The commercial robbery is distributed within certain regions of commercial areas while the other two kinds of robbery events occur more dispersedly in larger numerous grid regions.



(a) spatiotemporal distribution of commercial robbery



(b) spatiotemporal distribution of pedestrian robbery



(c) spatiotemporal distribution of residence robbery

**Figure 1. Spatiotemporal distributions of three kinds of robbery events in Atlanta, US.**

### 2.2 Data Organization

The data is organized as a three-dimensional tensor, and each dimension respectively represents the time, space, and number of crime occurrences within the spatiotemporal region. The space and time are firstly evenly divided into bins, and then the crime occurrence number in each space-time cell (bin/grid) are counted. Following the study of [5], we regard the number of crime occurrences in a central grid along with its eight neighborhood grids as the input of the LSTM, and predict the future crime occurrence in each central grid. Two factors may affect the predictability of the model, the cell size and the input history length. Generally, the smaller the cell size is, the harder to accurately forecast the future crime occurrences because of its randomness. The input historical length also exerts influences on model performance. For example, a crime may repeat with a larger time lag in a region, thus input of a short length historical data may fail the model to capture such long-time lag characteristics. The effects of the cell size and time length, or spatiotemporal scales, will be explored in the following study.

## 3. METHODOLOGY

Long-Short-Term-Memory (LSTM) method is used for crime prediction in this paper. LSTM is a variant of Recurrent Neural Networks (RNN) that was firstly proposed by Hochreiter & Schmidhuber (1997) [12] and recently improved and popularized by Alex Graves. RNN allows connections between nodes to form a directed cycle [6]. LSTM algorithm solves the problem of gradient disappearance of RNN and can well deal with the long-term dependence problem. LSTM has specially designed memory cells responsible for the host information, and the hidden layer updates the information through the memory cells. The cell state, which is determined by a structure called ‘gate’ that removes or adds information to the cell state, is the key to LSTM. Three gates are involved in LSTM, the input gate, the forgetting gate, and the output gate. The Cell state and information update is achieved by the following formula:

$$f_t = \sigma(W_{fx}x_t + W_{fh}h_{t-1} + b_f)$$

$$i_t = \sigma(W_{ix}x_t + W_{ih}h_{t-1} + b_i)$$

$$\begin{aligned}
o_t &= \sigma(W_{ox}x_t + W_{oh}h_{t-1} + b_o) \\
\tilde{c}_t &= \tanh(W_{cx}x_t + W_{ch}h_{t-1} + b_c) \\
c_t &= f_t c_{t-1} + i_t \tilde{c}_t \\
h_t &= o_t \tanh(c_t)
\end{aligned}$$

where  $f_t$ ,  $i_t$ ,  $o_t$  respectively are forgetting gate, input gate and output gate.  $W_{fx}$ ,  $W_{fh}$ ,  $W_{ix}$ ,  $W_{ih}$ ,  $W_{ox}$ ,  $W_{oh}$ ,  $W_{cx}$ ,  $W_{ch}$  is the coefficient matrix, and  $b_f$ ,  $b_i$ ,  $b_o$ ,  $b_c$  is the partial vector. The forgetting gate reads  $h_{t-1}$  and  $x_t$ , and outputs a value between 0 and 1 to each number in the cell state  $c_{t-1}$ . 1 means completely retained and 0 means completely discarded. The sigmoid layer, called the input gate layer, determines what values we will update. The tanh layer creates a new candidate value vector  $\tilde{c}_t$ . After updating the old cell state time, a sigmoid layer is run as the output gate to determine which parts of the cell state will be exported. The cell state was processed through tanh to obtain the final output  $h_t$ .

The time dynamics of the crime pattern have certain rules. For example, the number of crimes in the evening of a day may be different from that in the morning, and the number of crimes on weekends and working days of a week may also be different. If only the most recent time data is considered to predict, and the historical time data is ignored, then the predicted results will have a certain deviation. Secondly, the crimes studied here are different robbery crimes. In terms of geographical distribution, the criminal events in commercial areas, residential areas and pedestrians conform to the distribution in different regions. It is mentioned in 2.1 that the spatial distribution changes with certain regularity. Since LSTM has the advantage of long-term memory, it can effectively solve the problem of crime predictions with high long-time spatial dependency.

## 4. EXPERIMENTS

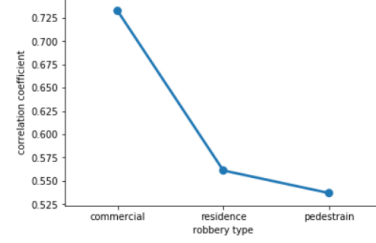
### 4.1 Experiment Design

Two groups of experiments are designed, respectively to compare the crime predictability in Atlanta and Baltimore, and each group contains three sub-experiments. In the three sub-experiments, the first one is to compare the predictability of different types of crimes under the premise of fixed cell size and time length. Firstly, different types of crimes have different degrees of spatial and temporal agglomeration. For example, some crimes are spatially concentrated. Secondly, different spatiotemporal scale settings may affect the results of the model. Therefore, we set up a second and third sub-experiments to explore the impact of different spatial and temporal granularity on different types of crimes. The correlation coefficient  $R$  between the predicted spatiotemporal crime occurrences and the real values is used to measure the model performance, and cross-validation is used to test the model with a ratio of training set and test set (4:1) [16].

### 4.2 Result and Analysis of Atlanta

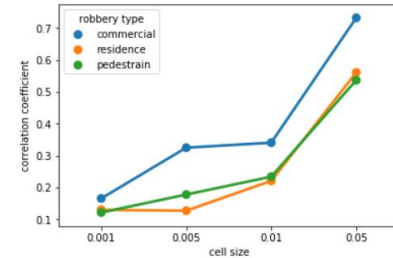
The first experiment is to explore the predictability of three robbery crimes using a fixed cell size and time length in Atlanta. The predictable result is shown in Fig.2. Here, we divide the space into 0.05 degree of grids, and the input time length is set to 30 days. It

can be found that the predictability of commercial robbery in this case is much higher than the predictability of the other two types of regional robbery crime, with a  $R$  value up to 0.73. The least predictable robbery is the pedestrian robbery. This indicates that the commercial robbery occurs more regularly, and more easily to be solved, while the pedestrian robbery occurs more randomly, and is harder to accurately forecast.



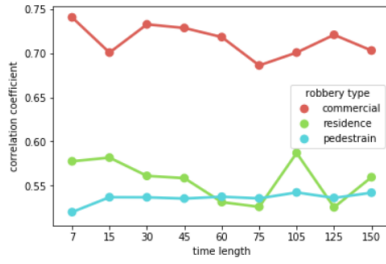
**Figure 2. The predictability comparison of three kinds of robbery events in Atlanta.**

The second experiment is to explore the influence of cell sizes on the predictability of the three robbery crimes in Atlanta. The comparison result is shown in Fig. 3. Here we set a fixed time length value of 30 days, and change the cell size values from 0.001 to 0.05. It can be found that when the cell size is 0.05, the predicted scores of the three robbery crimes all achieve the highest value. However, the score differently changes as the cell size becomes smaller. The predictability of commercial robbery events is more stable within the region of 0.005-0.01 cell sizes, indicating that this kind of robbery is more densely distributed in this region. While the corresponding distributions of the other two kinds of robbery events are more randomly. It can be concluded that too small space division is difficult to predict the spatial distribution of crime trends.



**Figure 3. The predictability comparison of three kinds of robbery events using different cell sizes in Atlanta.**

The third experiment is to explore the influence of different time lengths on the predictability of three robbery crimes in Atlanta. The comparison result is shown in Fig. 4. Here we set a fixed cell size of 0.05, and test varying time lengths from 7 days to 150 days. It can be found that the predictability of the three robbery crimes fluctuate differently as the time length changes. The three types of robbery also differ in the magnitude of their peaks in predictability. The predictability of commercial robbery is the best when the time length is 7 days. The predictability of robberies in residential and pedestrian areas peaked at 105 days. The predictability of commercial and residential robberies fluctuates greatly with the change of time length, while pedestrian robberies are relatively stable. It shows that there are differences among the three crime cycles.

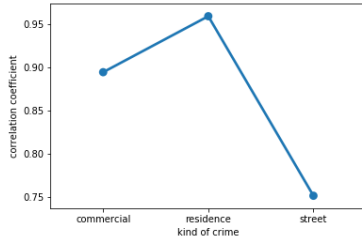


**Figure 4. The predictability comparison of three kinds of robbery events using different time lengths in Atlanta.**

### 4.3 Result and Analysis of Baltimore

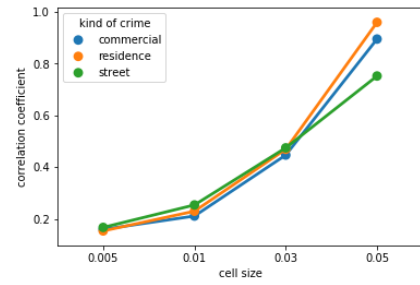
In 4.2, we have used the model to predict the Atlanta crime data set. In this section we compare the three kinds of robbery crimes in Baltimore. The experimental design is the same as the previous part.

The first experiment is to compare the predictability of three robbery crimes in Baltimore under fixed cell size and time length. We still set the cell size at 0.05 and the time length at 30 days. The comparison results are shown in fig.5. It can be found that the crime predictability of residential area, and commercial area in this data set are significantly higher than that of street (pedestrian robbery). It demonstrates that the geographical location and time of crime in the residential and commercial areas in this city are relatively fixed, and more predictable.



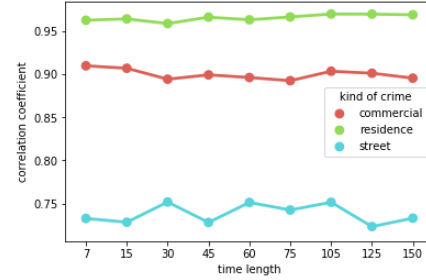
**Figure 5. The predictability comparison of three kinds of robbery events in Baltimore.**

The second experiment compared the effects of different cell sizes on the predictability of three robbery crimes in Baltimore. The comparison results are shown in Fig. 6. We still set a fixed time length of 30 days and change the cell size range from 0.005 to 0.05. The same conclusion as the previous part can be drawn that the three crimes are most predictable at the maximum cell size when the cell size is 0.05. The predictability of residential crime events in this data set is slightly higher than that of the commercial crime, and pedestrian crime events are much less predictable than the other two robbery events when the cell size equals to 0.05. While the predictability of these three kinds of robbery events show similar predictability when the cell size is smaller. Experiments indicate that the residential and commercial crime events are more regularly distributed at a large spatial scale, while the pedestrian robbery events are much more randomly distributed at all spatial scales.



**Figure 6. The predictability comparison of three kinds of robbery events using different cell sizes in Baltimore.**

The third experiment compared the effects of different time length changes on the three crime predictors under the premise of fixed cell size. We repeat the steps and settings of the previous experiments, the comparison results show that residence increases very slowly and slightly with the change of time length in Fig.7. Commercial districts are most predictable when time length is 7 days. The street area peaked in predictability at time length of 30 days. The predictability of commercial and residence area fluctuates less with time length, while street fluctuates more obviously.



**Figure 7. The predictability comparison of three kinds of robbery events using different time lengths in Baltimore.**

## 5. CONCLUSION

Crime prediction is a complex and challenging task, and accurately forecast the spatial and temporal distribution trend of crime is one of the key problems. This paper discusses the predictability of LSTM to different kinds of robbery crimes, and explores the influence of multiple space-time scales on the prediction. We find that the three kinds of robbery events have different predictability between different cities and categories. In Atlanta, the commercial robbery events are more predictable than pedestrian robbery and residence robbery events, and the pedestrian robbery events are hardest to forecast because of its dispersedly and randomly distributed. While in Baltimore, residential robbery events are most predictable. Experiments of two cities also show similar results. As the grid size or cell size decreases, the predictability of the three robbery events decreases consequently. The predictability of the three robbery events fluctuates differently as the time length changes, reflecting three different crime periodical cycles or time dependencies.

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